30/7/26

Experiment - 2

Study of Different types of Network Cables

Aim:

To study of different types of network cable

Different type of cables used in networking are:

1. Unshielded Twisted pair (UTP) cable.
2. Shielded Twisted pair (STP) cable.
3. Coaxial cable
4. Fibre optic cable.

1. Cable Type:

UTP.

Category & Maximum Data Transmission

Category 3 - 10bps

Category 5 - Upto 100 Mbps

Category 5e - 1Gbps.

Advantages:

* Cheaper in cost
* Easy to install as they have a smaller overall diameter

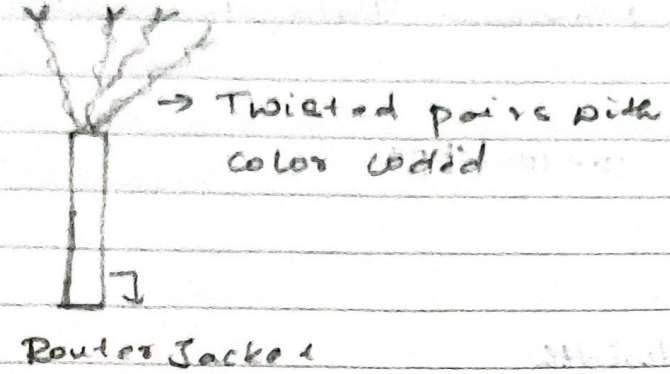
Disadvantages

* More prone to (EMI) Electromagnetic interference and noise

Application / Use.

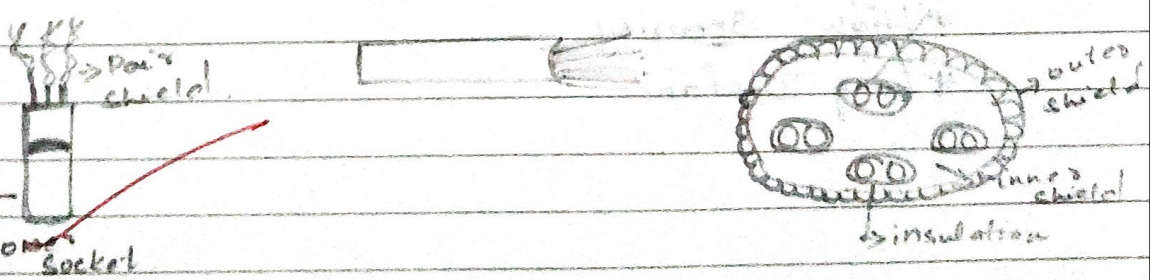
* 10 Base - T Ethernet
* Fast Ethernet. Gigabit Ethernet.
*

Image:



2. Cable Type:
STP, SSTP
Category & Maximum Data Transmission
Category 6, 6a - 10Gbps
Category 7 - 10Gbps.
Advantages:
* Shielded
* Faster than UTP
* Less susceptible to noise & interference.
Disadvantages:
* Expensive
* Greater installation effort.
Application / Use
* Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres.
* Gigabit Ethernet 10G Ethernet (100m).

Image.



3. Cable Type:
Coaxial Cable
Category & Maximum Data Transmission

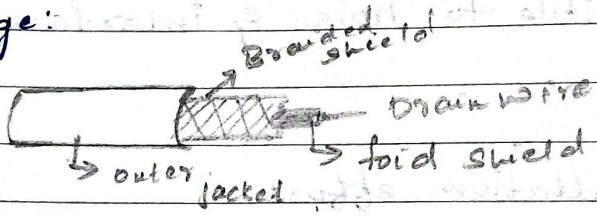
RG-6
RG-59 - 10-100 Mbps
RG-11

Advantages:
* High bandwidth
* Immune to interference
* Versatile.

Disadvantages:
* Limited Distance
* Cost
* Size is bulky.

Application / Use.
Speed of signal in Room Television network
High speed internet connections.

Image:



4. Cable Type:
Fibre optics:
Category & Maximum Data Transmission.
Single Mode, Multimode - 100Gbps
Advantages:
* High Speed
* High bandwidth
* High Security
* long Distance.

Disadvantages:

* Expensive.
* Requires skilled installers.

Application / Use

Maximum distance of fibre optics cable is around 100 meters.

Images:

Result:

Thus the study of different types of network cable has been studied successfully.